LETTER

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Deposition of 1.88-billion-year-old iron formations as a consequence of rapid crustal growth

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Iron formations are chemical sedimentary rocks comprising layers of iron-rich and silica-rich minerals whose deposition requires anoxic and iron-rich (ferruginous) sea water. Their demise after the rise in atmospheric oxygen by 2.32 billion years (Gyr) ago[1] has been attributed to the removal of dissolved iron through progressive oxidation[2] or sulphidation[3,4] of the deep ocean. Therefore, a sudden return of voluminous iron formations nearly 500 million years later poses an apparent conundrum[3,5]. Most late Palaeoproterozoic iron formations are about 1.88 Gyr old[6–8] and occur in the Superior region of North America[5,9,10]. Major iron formations are also preserved in Australia, but these were apparently deposited[11] after the transition to a sulphidic ocean at 1.84 Gyr ago that should have terminated iron formation deposition[4], implying that they reflect local marine conditions[5,12]. Here we date zircons in tuff layers to show that iron formations in the Frere Formation of Western Australia are about 1.88 Gyr old, indicating that the deposition of iron formations from two disparate cratons was coeval and probably reflects global ocean chemistry. The sudden reappearance of major iron formations at 1.88 Gyr ago—contemporaneous with peaks in global mafic–ultramafic magmatism[13,14], juvenile continental and oceanic crust formation[15,16], mantle depletion[17,18] and volcanogenic massive sulphide formation[5,19]—suggests deposition of iron formations as a consequence of major mantle activity and rapid crustal growth[5,10,15,20]. Our findings support the idea that enhanced submarine volcanism and hydrothermal activity linked to a peak in mantle melting released large volumes of ferrous iron and other reductants that overwhelmed the sulphate and oxygen reservoirs of the ocean, decoupling atmospheric and seawater redox states, and causing the return of widespread ferruginous conditions. Iron formations formed on clastic-starved coastal shelves where dissolved iron upwelled and mixed with oxygenated surface water. The disappearance of iron formations after this event may reflect waning mafic–ultramafic magmatism and a diminished flux of hydrothermal iron relative to seawater oxidants.

Iron formations are chemical sedimentary rocks whose deposition requires an anoxic water column to transport ferrous iron to sites of iron precipitation. Their formation was episodic and broadly coeval with, and in some cases possibly linked to, periods of mantle plume activity as recorded by peaks in mafic and ultramafic igneous activity[5,10]. The largest accumulations of iron formation were deposited between 2.7 and 2.45 Gyr ago[5,10], before the rise of atmospheric oxygen by 2.32 Gyr ago[1]. The decline of iron formations after 2.45 Gyr ago is interpreted to reflect the removal of ferrous iron by progressive oxidation of the deep ocean[2] or, alternatively, by the precipitation of iron sulphides due to enhanced riverine sulphate delivery to the oceans after the rise of atmospheric oxygen[3,4]. Important exceptions are late Palaeoproterozoic iron formations from the Superior craton of North America and the Frere Formation of the Earaheedy Group, Western Australia[5,9,10] (Fig. 1). These late Palaeoproterozoic iron formations comprise very thick and extensive sequences of granular iron formation that were deposited in shallow coastal environments close to the storm and fair-weather wave base[5]. Banded iron formations are also present and indicate deposition in deeper water settings. The rare-earth element chemistry of these iron formations indicates a significant hydrothermal input[11,21] and the existence of a redox-stratified ocean during their deposition[22].

Iron formations in North America were deposited at 1.88 Gyr ago[6–8] and can be traced for about 3,000 km along the Superior craton (Fig. 1). The age of the granular iron formation of the Frere Formation is poorly constrained, but is considered to be between 1.84–1.65 Gyr ago[11,23], coeval with the development of euxinic conditions in the global ocean linked to the demise of Superior-type iron formations[3,4]. If the deposition of the granular iron formations in Australia, which required ferruginous conditions, was synchronous with sulphidic conditions in North America, then the late Palaeoproterozoic iron formations may reflect the chemistry of local, restricted marine basins and therefore provide no information about the redox state of the global ocean[5,12].

To better constrain the age of the Australian granular iron formations, we sampled tuffaceous and volcaniclastic layers in the basal Frere Formation from a diamond drill-hole (TDH26) drilled on the southwestern margin of the Earaheedy basin, Western Australia (Fig. 1; Supplementary Fig. 1). The Frere Formation is a succession 600–1,200 m thick of granular iron formation, green shale and ferruginous siltstone with minor carbonate, chert and banded iron formation, which is bounded by carbonate units of the lower Earaheedy Group. The occurrence of oolitic and stromatolitic carbonates and the range of sedimentary structures indicate a shallow marine environment[11].

Three samples from the base of the Frere Formation (TDH26, 343.47–343.53 m; 374.45–374.55 m and 386.38–386.46 m) preserve thin tuff beds comprising angular fragments of volcanic quartz, euhedral zircon and monazite in a matrix of chlorite. Zircons extracted from the tuff layers were analysed by sensitive high-resolution ion microprobe (SHRIMP) using established techniques[24]. All zircon U–Pb data are given in Supplementary Tables 1, 2 and 3. Zircon crystals in the lower tuff bed, which occurs about a metre above the base of the Frere Formation, yielded a weighted mean $^{207}Pb/^{206}Pb$ age of 1,891 ± 8 million years (Myr) (mean square weighted deviation, MSWD = 1.5; $n = 28$) (see Supplementary Figs 5–8). Zircons in the middle tuff bed gave a $^{207}Pb/^{206}Pb$ age of 1,890 ± 10 Myr (MSWD = 1.3; $n = 15$) and zircons in the upper tuff gave 1,885 ± 18 Myr (MSWD = 1.1; $n = 9$) (see Supplementary Figs 9 and 10).

The zircons are interpreted to record the age of deposition synchronous with volcanism on the basis of their euhedral shape, igneous growth zoning and uniform size (mostly between 10 and 30 μm), and the presence of pyroclastic debris and angular slivers of quartz in the tuff beds. The 1.89-Gyr depositional age is consistent with the age of the youngest detrital zircons in the lower Earaheedy Group (about 2.0 Gyr ago)[11]. In view of our new age constraints, previous interpretations for the maximum age of the Earaheedy Group (<1.84 Gyr

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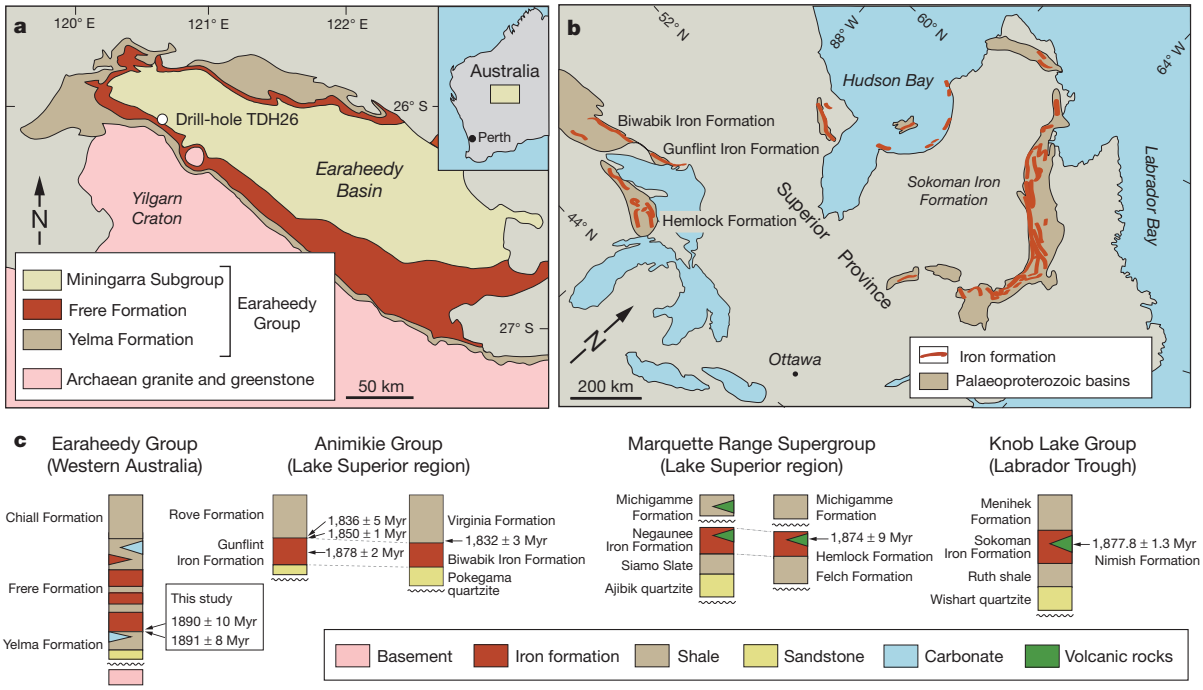


Figure 1 | Location and stratigraphy of late Palaeoproterozoic iron formations. Maps show the distribution of the main Late Palaeoproterozoic iron formations in Western Australia (**a**), and North America (**b**). **c**, Stratigraphic columns of successions containing Late Palaeoproterozoic iron formations and their age constraints (after refs 5–9, 11 and 30).

ago)[11,23] are invalid and the underlying Yerrida Group must be older than 1.89 Gyr ago.

The 1.89 Gyr age of the basal Frere Formation closely corresponds with the age of iron formations in the Superior craton (1.88 Gyr ago)[6–8]. Palaeomagnetic reconstructions suggest that Western Australia and North America were far apart at this time[14], supporting arguments that these iron formations were deposited on the margins of disparate cratons. Therefore, their deposition is likely to reflect the chemistry of the global ocean rather than restricted basin-scale conditions.

The deposition of vast volumes of iron formation on continental shelves indicates that the deep oceans were anoxic, non-sulphidic and rich in dissolved iron 1.88 Gyr ago. The sudden reappearance of these conditions after an interval of over 500 Myr is surprising because iron is inferred to have been progressively removed from the ocean after the rise in atmospheric oxygen between 2.47 and 2.32 Gyr ago[1–3]. After this time, the atmosphere–ocean system shifted to a new redox state where ocean chemistry was increasingly influenced by the high flux of oxygen and sulphate from surface environments and atmosphere. The return to ferruginous, Archaean-like conditions in the ocean was apparently not accompanied by a fall in atmospheric oxygen levels, at least not to levels before the Great Oxidation Event, as indicated by mass-independent fractionation of sulphur isotopes in sedimentary sulphides[1,25] (Fig. 2). This apparent decoupling of the redox state of the atmosphere and the deep ocean at 1.88 Gyr ago is thus problematic.

A possible explanation for the presence of voluminous iron formations at 1.88 Gyr ago is offered by their deposition during a period of intense igneous activity (Fig. 2a, c and d). For example, between 1.89 and 1.87 Gyr ago, extensive mafic and ultramafic magmatism occurred across the world, including the emplacement of dyke swarms and sills with mineralization of Ni, Cu and the platinum group elements, and locally, basaltic flows, in the Superior, Wyoming and Slave cratons in North America, the Dharwar and Bastar cratons in India, the Siberian craton, the Kaapvaal and Zimbabwe cratons in southern Africa, and Baltica[13,14]. The emplacement of numerous large igneous provinces at that time might reflect the activity of multiple mantle plumes (a superplume event)[10,15,20].

The emplacement of multiple large igneous provinces at 1.88 Gyr ago corresponds with a major episode in continental and oceanic crustal growth recorded by emplacement ages of juvenile igneous rocks[15,20] (Fig. 2c) and the coupled Lu–Hf and O isotope composition of detrital zircons[16] (Fig. 2d). The peak 1.88 Gyr ago in juvenile crust formation is coeval with a major peak in mantle depletion ages, suggesting that new continental crust generation was linked with large-scale mantle melting[17,18]. The peak in crustal growth at that time also corresponds with supercontinent assembly.

The inferred geological consequences of this mantle superplume event include increased rates of seafloor spreading, the generation of seamounts and oceanic plateaus, enhanced ocean anoxia and marine transgressions[5,10,20]. Widespread mafic–ultramafic magmatism and enhanced oceanic crust production is also interpreted to enhance submarine volcanism and hydrothermal activity and the influx of vast volumes of metals and metalloids, including base metals and iron into the deep ocean[10,15,20,26]. There is strong evidence for widespread back-arc magmatism and hydrothermal activity at 1.88 Gyr ago, with a major peak in the formation of volcanogenic massive sulphide (VMS) deposits[5,19] (Fig. 2b). The metals for VMS deposits are released by hydrothermal leaching of volcanic rocks, whereas the sulphur in hydrothermal systems is derived from the reduction of seawater sulphate and leaching of mafic volcanic rocks.

The hydrothermal systems that generate VMS deposits also emit large volumes of dissolved iron into the deep ocean. Under anoxic, sulphidic (euxinic) conditions, as have been inferred to exist in global oceans after and possibly before 1.88 Gyr ago[3,4], hydrothermal iron would rapidly precipitate as iron sulphides. If the flux of sulphate was higher than that of dissolved iron, iron would have been titrated from the ocean[3,4]. However, the presence of vast accumulations of shallow-water granular iron formation at 1.88 Gyr ago indicates that dissolved iron accumulated in the deep ocean and periodically upwelled onto the coastal zones of continental shelves, where it was precipitated as iron-oxyhydroxide particles[22]. The large-scale transport of dissolved iron from the deep ocean to the shallow sea implies that most of the water column was ferruginous 1.88 Gyr ago.

Recent iron speciation data for Proterozoic shales suggest that the deep ocean was at a very low oxidation state with locally developed ferruginous and euxinic conditions, below oxygenated surface water[27] (Fig. 2h). Similarly, iron and sulphur data from the 1.88–1.83-Gyr-old

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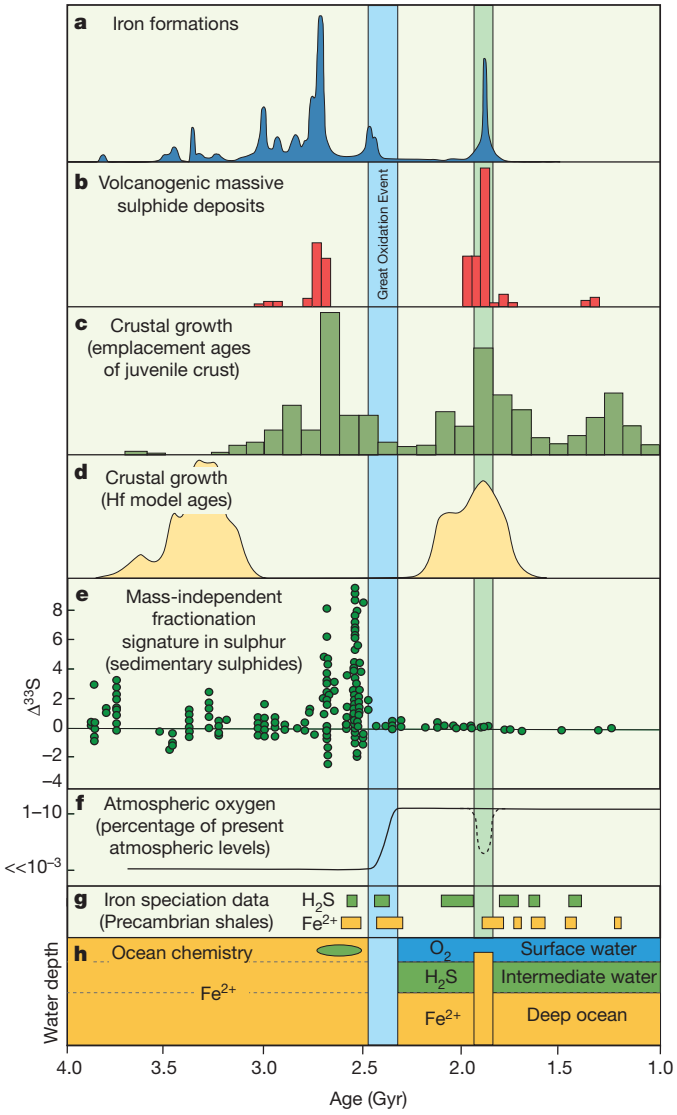


Figure 2 | Summary of the age distribution of iron formations, VMS deposits, juvenile crust and ocean chemistry for the Precambrian. **a**, Secular trend in the distribution of iron formations (after ref. 10). **b**, The distribution of VMS deposits over time (after ref. 5). **c**, Distribution of U–Pb zircon ages for juvenile crust (after ref. 15). **d**, Gaussian probability distribution of Hf model ages for zircons with low $\delta^{18}O$ values (after ref. 16). **e**, Plot of $\Delta^{33}S$ values versus sample age (after ref. 25). **f**, Summary of the atmospheric oxygen concentrations (after ref. 27). **g**, Iron speciation record of mudrocks indicating ferruginous and sulphidic conditions in the Proterozoic oceans (after ref. 27). **h**, A summary of the inferred ocean chemistry between 4.0 and 1.0 Gyr ago. After the Great Oxidation Event, the chemistry of the oceans shifted to a non-steady redox state controlled by fluctuating fluxes of oxygen, sulphate and iron, where the redox boundaries among oxic, sulphidic and ferruginous conditions were dynamic and moved vertically and laterally in response to variations in submarine hydrothermal fluxes and surface oxygenation.

Animikie basin in the Superior region of North America suggest that ferruginous conditions dominated during deposition of the 1.88-Gyr-old Gunflint Formation and that euxinic conditions subsequently developed at mid-depths along palaeoshorelines, where dissolved iron was removed from the water column, terminating the deposition of iron formations[28]. Although the development of an anoxic water column is required for the mobility of dissolved iron, the coexistence of locally developed euxinic conditions has been inferred to prevent the deposition of iron formations after about 1.88 Gyr ago[28].

We propose that the apparent transition to a ferruginous water column (including mid-depths) at 1.88 Gyr ago was triggered by the emanation of vast volumes of ferrous iron and other reduced species (such as H_2, CH_4 and Mn^{2+}) during a short-lived period of intense submarine volcanism and hydrothermal activity. A peak in the accumulation of VMS deposits at 1.88 Gyr ago[5,19] shows that large volumes of hydrothermal iron were emitted and subsequently deposited as iron sulphides near vents, removing sulphur compounds from sea water. The concentration of seawater oxygen and sulphate at about 1.88 Gyr ago was significantly lower than that of today[3,4], so that the enhanced influx of reduced species delivered by hydrothermal plumes to the oceans may have overwhelmed both oxidants in sea water[29], greatly reducing the spatial extent of euxinic and oxygenated zones in the oceans.

Our findings provide a simple explanation for the apparent conundrum of the sudden reappearance of iron formations after the first major rise in atmospheric oxygen. Specifically, mantle-driven processes at about 1.88 Gyr ago not only acted as a source of dissolved iron, but also modified ocean chemistry and redox state by releasing a large flux of reductants that temporarily overwhelmed the supply of oxidants to the deep ocean from shallow waters and atmosphere. This led to the development of a largely ferruginous water column beneath an oxygenated surface zone (Fig. 2h). Iron-oxyhydroxides accumulated on the sea floor in shallow marine, coastal environments, where Fe^{2+} was oxidized in clastic-starved settings, forming extensive deposits of granular iron formation.

Although the chemistry of the deep ocean apparently returned to Archaean-like conditions at 1.88 Gyr ago, atmospheric oxygen apparently did not fall to pre-Great Oxidation Event levels (Fig. 2e, f). The decoupling between the redox state of the atmosphere and ocean may be explained by the release of reductants into the bottom of the ocean, where they were buffered by the seawater oxidants, largely restricting anoxia to the ocean. Our findings also provide a mechanism for the cessation of iron formation deposition. Rather than being a consequence of changing surface conditions, we propose that the demise of iron formations corresponds with the termination of this short-lived interval of global mantle-driven magmatism and crustal growth. After 1.88 Gyr ago, the deposition of iron formations was no longer favoured because the flux of oxidants from the atmosphere and surface ocean significantly exceeded the long-term rate of delivery of hydrothermal Fe and reductants to the deeper parts of the oceans. Major iron formations did not reappear until the late Neoproterozoic era, when 'snowball Earth' conditions radically changed the chemistry of the global ocean and once again favoured the deposition of iron formations.

METHODS SUMMARY

Three tuff beds were identified in the basal Frere Formation, Earaheedy Group, Western Australia, from drill-hole TDH26, at depths of 386.38–386.46 m, 374.45–374.55 m and 343.47–343.53 m (see Supplementary Fig. 1). Polished thin sections of the tuffs were examined using transmitted light and reflected light microscopy. Zircon crystals (typically between 10 and 30 μm in size) were identified using their characteristic optical properties: shape, straight extinction, oscillatory zoning, high refractive indices and high birefringence. Zircons that were sufficiently large for analysis by ion microprobe (≥15 μm) were drilled out of polished thin sections in ~3-mm discs and set in conventional 25-mm epoxy resin mounts (see ref. 24). Chips of the zircon Pb/U reference standard (BR266) were cast in separate epoxy resin mounts. *In situ* U–Th–Pb analysis of zircon was carried out using a SHRIMP at the John de Laeter Centre for Isotope Research, Curtin University, Perth, Australia. All data were recorded in two 2-day analytical sessions. In the first session, the primary ion beam was ~0.5 nA on a ~15-μm spot and Pb/U external precision was 1.1% ($n = 19$). In the second session, the primary ion beam was ~0.3 nA on a ~12-μm spot and Pb/U external precision was 1.3% ($n = 34$). No corrections were required for instrumental mass fractionation of Pb isotopes. Following SHRIMP geochronology, zircon crystals were examined using a JEOL-6400 scanning electron microscope located in the Centre for Microscopy, Characterisation and Analysis at the University of Western Australia. Details of the sample description, SHRIMP analytical methods, SHRIMP data tables and data assessments are presented in the Supplementary Information.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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